HridayTwin
by MyZenn

HridayTwin: a stress-aware digital twin for hypertension

Predicts a blood-pressure surge 24 hours before it happens, explains why, and simulates which intervention would prevent it. Built on MyZenn.

Predict
24 hours ahead

Explain
personal drivers

Simulate
what will help

Prevent
with MyZenn



Team AlphaGo | Jadavpur University | Happiest Health Digital Twin Challenge 2026 | github.com/rohanjain1648/HridayTwin

India’s hypertension crisis happens between clinic visits.

188 million
Indian adults have hypertension
(WHO, 2023)



have it under control
(WHO, 2023)

Blood pressure is checked every few months; **surges happen on ordinary days in between.**

Everyday triggers: stress, short or broken sleep, missed doses, salty meals.

These show up on a smartwatch and in a wellness app **before they show up on the cuff.**

Doctors have no early warning and no way to test “what if” for one specific patient.



Clinic Visit 1
BP: **128/82**
Normal

90 days
No visibility



Clinic Visit 2
BP: **162/98**
High





Change detected early on smartwatch and wellness app

No early warning reaches the doctor

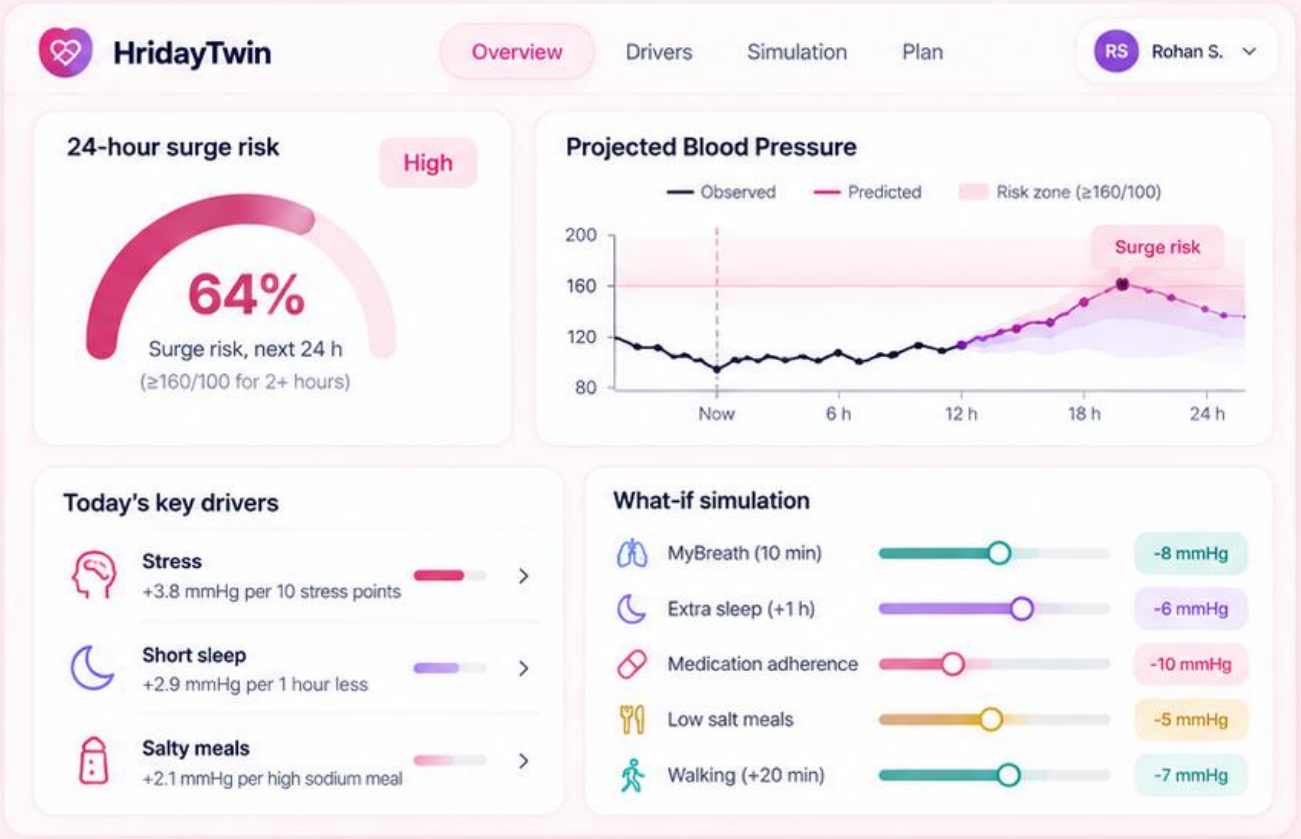


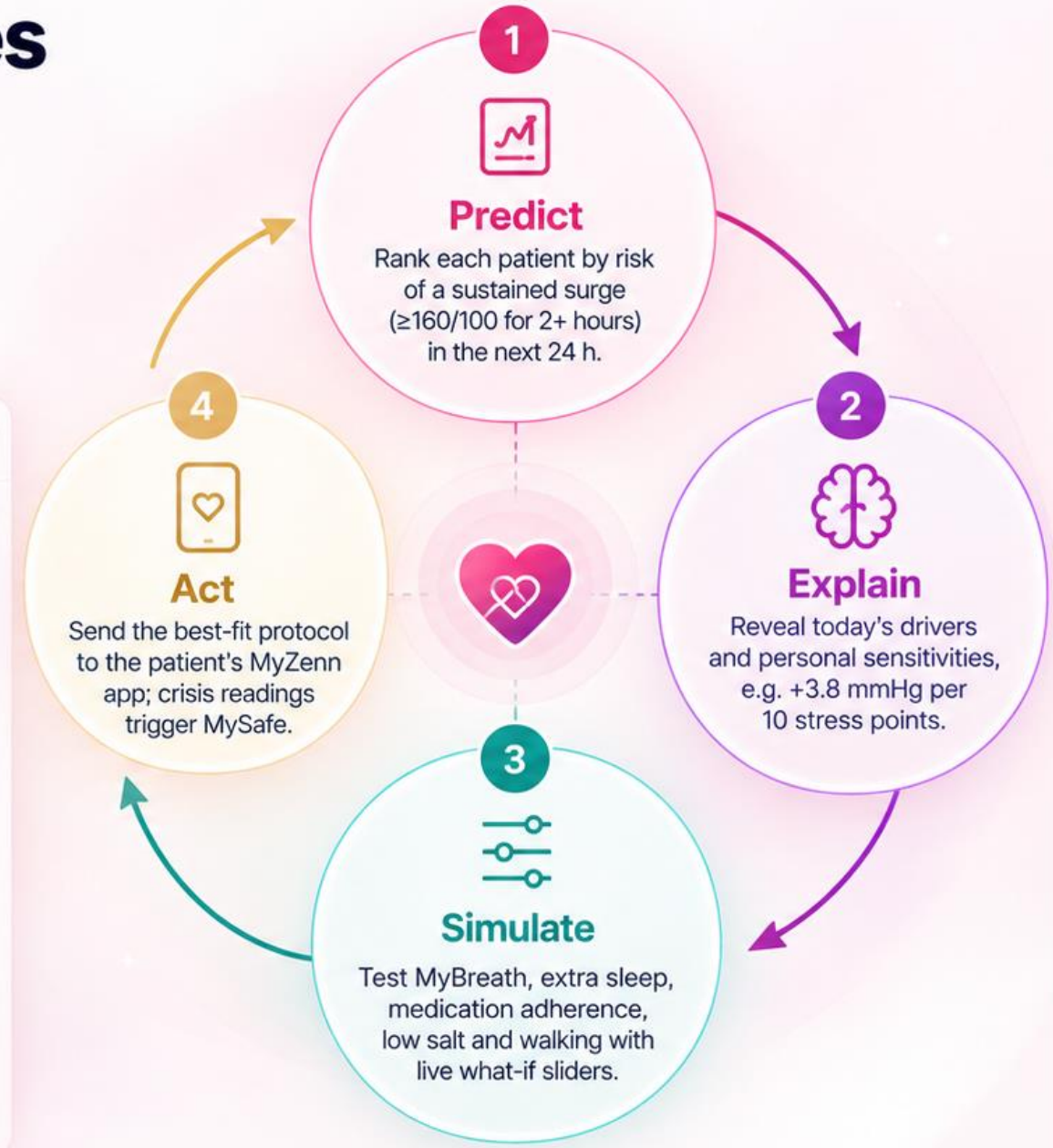
No way to test “what if” for this patient

Surges announce themselves in the mind and body first.

HridayTwin listens.

From early signals to personalized action — a prediction, prevention and intervention loop for hypertension.





DATA FOUNDATION

Two data streams, fused.
No real patient data anywhere.

A causal physiology simulator that combines static EHR with real-world wearable data to generate a realistic, explainable synthetic cohort for hypertension.

Static / historical EHR
(Synthea-style)

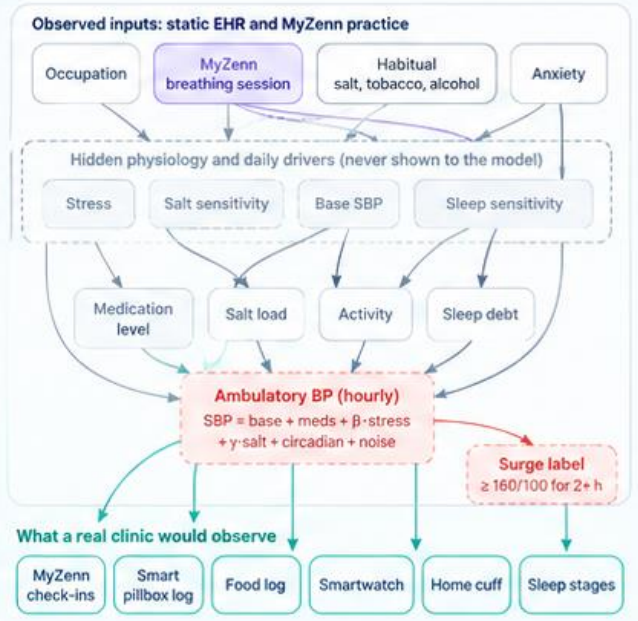
- Demographics, lifestyle, habitual salt
- Diagnoses in SNOMED-CT (T2DM, CKD, OSA, anxiety…)
- Labs in LOINC (eGFR, HbA1c, lipids, Na, K, UACR)
- Medications
- Genetic markers: AGT, ACE, CYP11B2, GNB3, polygenic score

Causal physiology simulator

Dynamic wearable + IoT
(hourly)

- Heart rate, HRV (RMSSD), steps, stress score
- Sleep stages and efficiency
- Home BP cuff (morning / evening, often skipped)
- Smart pillbox dose log
- MyZenn mood check-ins, breathing minutes, food log

Known causal ground truth enables explanation checks



400 patients × 60 days | **576,000** hourly rows | **4,577** surges | **Hidden ambulatory BP labels only** — never an input | Synthetic cohort • DPDP Act & HIPAA-aligned design

From raw streams to a doctor's decision in one pipeline.

A causal, hybrid digital twin system that turns multi-modal data into personalized risk, **explanations** and **actionable care.**

1 Data streams (synthetic + live)

Multi-modal inputs to build a realistic patient twin

Live sources
WebSocket replay (real-time app)
Webcam rPPG (camera vitals)
Bluetooth HR strap / smartwatch

MyZenn app signals
Mood check-ins, breathing minutes, salty-meal food log

Wearables + IoT (hourly)
Heart rate, HRV (RMSSD), steps, stress score, sleep stages, home BP cuff, smart pillbox log

Static / historical EHR (Synthea-style)
Demographics, lifestyle, habitual salt
Diagnoses in SNOMED-CT (T2DM, CKD, ...)
Labs in LOINC (eGFR, HbA1c, lipids, Na, K, UACR)
Medications
Genetic markers (AGT, ACE, CYP11B2, GNB3, PRS)

Physiology simulator
400 patients × 60 days
Causal model to generate realistic dynamics

Hidden ambulatory BP (labels only)
Used only for training/evaluation, never as a model input

2 Hybrid digital twin engine (Python)

Fuse data, model physiology, predict risk and recommend actions

Fusion feature store
71 strictly causal features
- aligned, cleaned and encoded
- no future leakage
- static + time-varying features

Mechanistic twin
Personal sensitivities, physiology and phenotype
24h BP projection under current conditions

What-if simulator
Change sleep, medication, salt, breathing, activity, etc.
Re-simulate 24h BP and risk

Risk engine (ML)
LightGBM with clinical monotone constraints
Calibrated surge probability
SHAP driver explanations

MySentinel policy
Contextual bandit selects personalized intervention (MyZenn protocol, MySafe) using risk, drivers and context

Labels (not an input)

3 Serving and experiences

From model outputs to real-world impact

MyZenn patient app
myzenn.com
- Personalized protocol
- Daily guidance and what-if
- MySafe crisis escalation
- Logs outcomes and adherence

Clinician console
- Risk dial and 24h BP projection
- Why this risk (SHAP explanations)
- What-if simulation
- Care plan and intervention send
- Track outcomes over time

FastAPI service
- REST APIs for risk, what-if, policy
- WebSocket for real-time data
- Secure, scalable deployment

Outcome feedback

Personalized protocol

Data streams
Multi-modal synthetic + live data and causal simulator

Hybrid twin engine
Fusion, mechanistic model, risk prediction, what-if simulation and intervention policy

Serving & feedback
Doctor and patient experiences with continuous learning

Runs locally with one command • Single Docker container deployment

Machine learning for accuracy, physiology for 'what if'.

A scenario edits recent behaviour; physiology moves blood pressure; ML re-scores risk.

Risk engine (ML)

LightGBM • 71 strictly causal features • Every 6 hours • 36 clinical monotonicity constraints • Platt-calibrated risk • TreeSHAP explanations

Risk prediction (next 24 h)



Top drivers (example)

High stress	+18%
Low sleep (4.5 h)	+12%
Missed dose	+11%
High salt intake	+8%
Low HRV	+7%

More medication adherence must never raise predicted risk; lower HRV must never lower it.

Shared patient state →

← Updated features & risk re-score

Mechanistic twin (physiology)

Bayesian posterior calibrated on the patient's own cuff readings

BP = base − meds·adherence + β·stress + γ·sleep debt + salt sensitivity·salt + circadian

Outputs

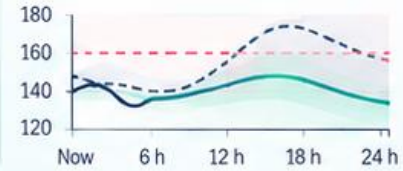
Personal sensitivities

e.g. mmHg per 10 stress points, per hour of sleep debt, per gram of salt; drug effect

Patient phenotype

stress-reactive, salt-sensitive, adherence-dependent, sleep-linked

24-hour BP projection with 80% uncertainty band



1 Doctor changes scenario sliders

- MyBreath minutes
- Extra sleep hours
- Every dose
- Lower salt intake
- More steps

→

Edit recent streams

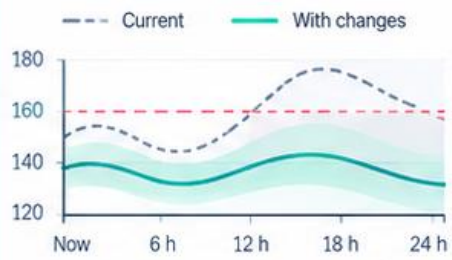
Update the patient's recent data

- MyBreath
- Sleep
- Doses
- Salt
- Steps

→

Physiology recalculates BP

Mechanistic twin simulates new BP



→

ML re-scores risk

Updated features → new risk



64% → 17%
with adherence rescue
(illustrative example)

→

Ranked care plan to MyZenn

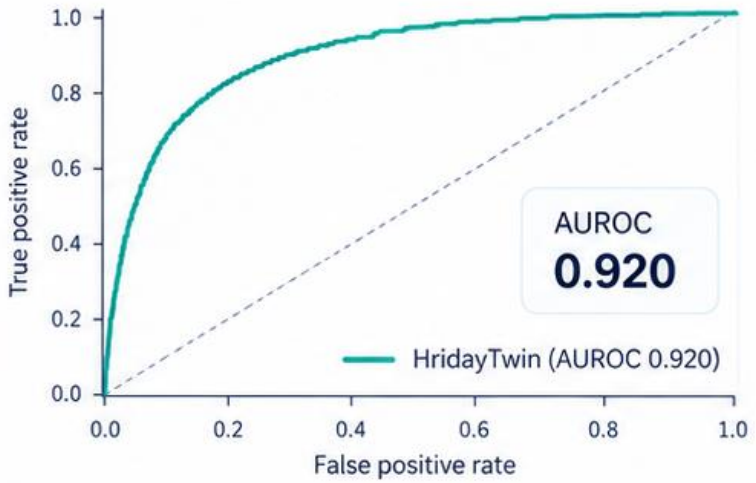
1. Adherence support (every dose)
2. MyBreath (10 min/day)
3. +1 hour sleep
4. Low salt meals
5. +20 min walking

A twin, not just dashboard: ML estimates risk; physiology tests counterfactuals.

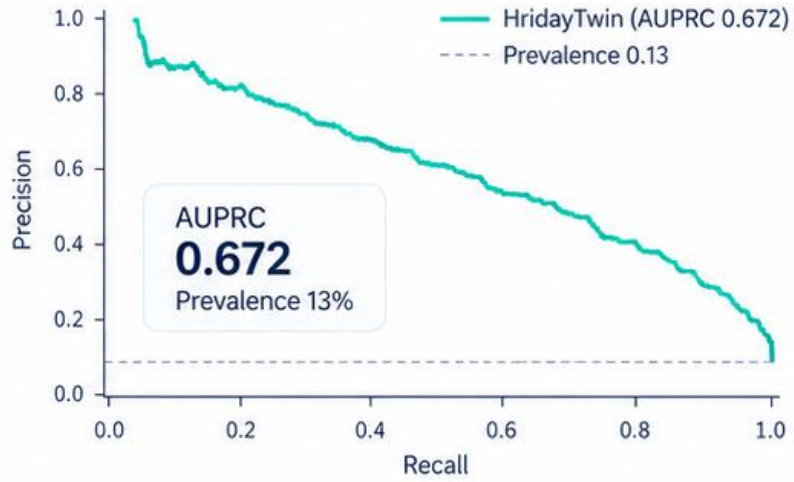
On 100 patients the model never saw, HridayTwin warns 19 hours ahead.

Early warnings give patients and doctors time to act.

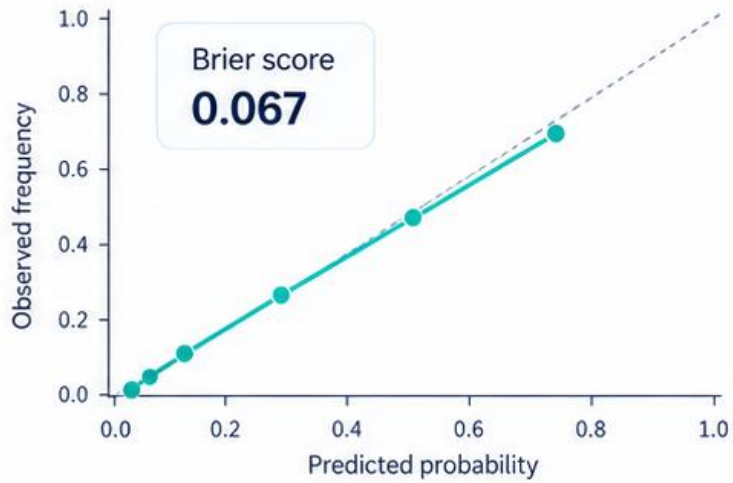
ROC curve (held-out patients)



Precision–Recall curve



Calibration plot (Brier: 0.067)



0.920
AUROC

AUPRC 0.672 · Brier 0.067

79%
of surges flagged in advance at 30% threshold

19 h
median warning before onset · 96% flagged ≥6 h ahead

0.6
high-tier alerts / patient / week · 97% specificity · 72% PPV

240 train · **60** validation · **100** held-out test patients

Patient-level separation prevents the same patient appearing in train and test.

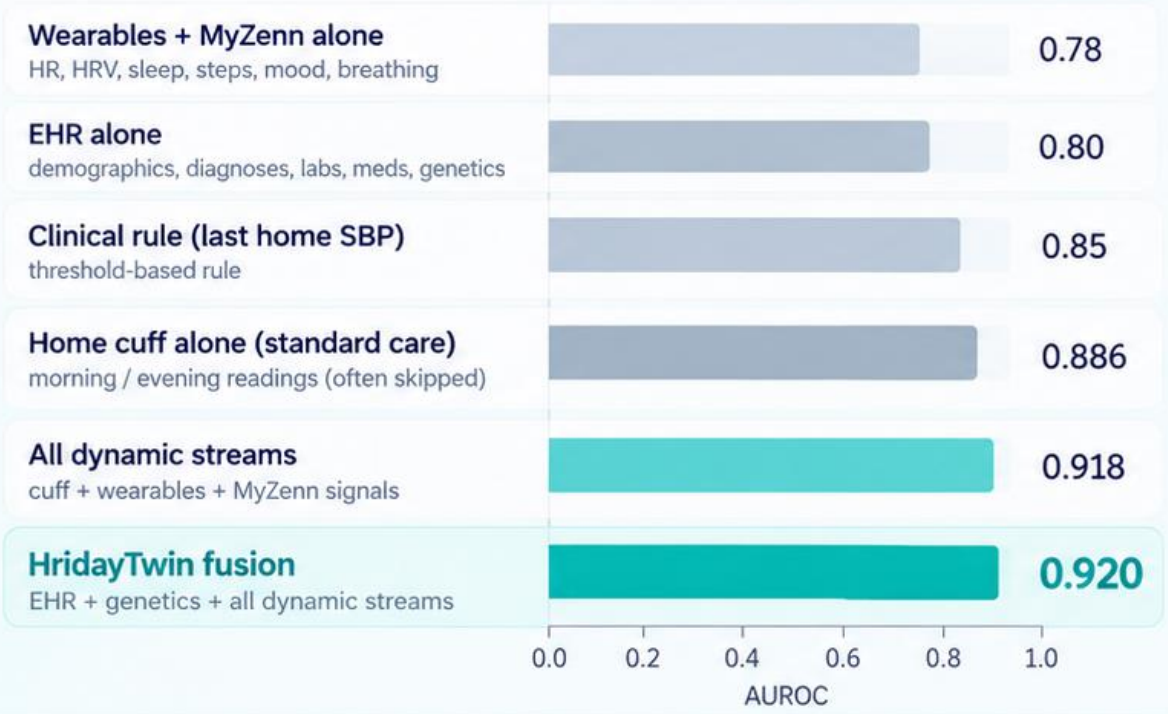
Alert burden matters: just **0.6** high-tier alerts per patient per week.

Every stream adds signal.
The twin recovers each patient's physiology.

HridayTwin
From signals to healthier days.

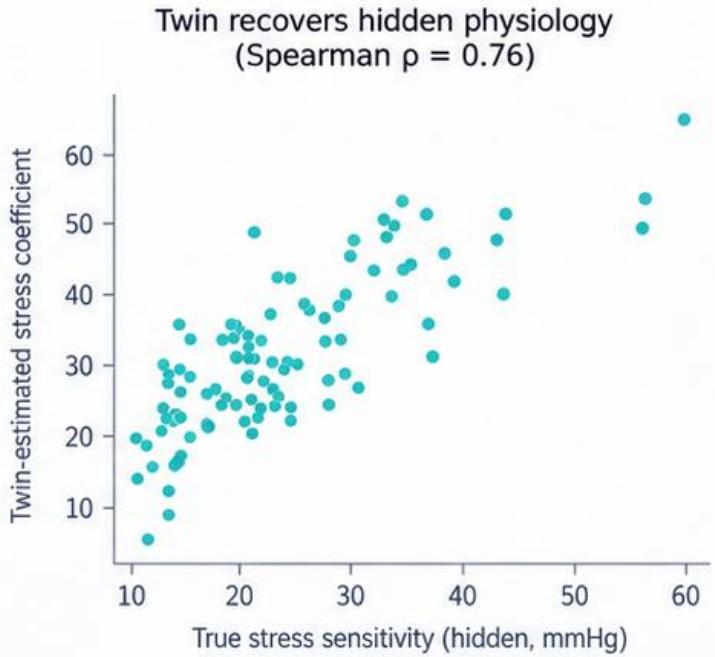
Fusion ablation · AUROC

Adding more streams consistently improves performance, with full fusion performing best.



Twin validation

The twin recovers hidden physiological sensitivities for each patient.



Recovered sensitivities correlate with ground truth

Stress sensitivity	ρ = 0.76
Drug effect	ρ = 0.59
Salt sensitivity	ρ = 0.47
Sleep sensitivity	ρ = 0.45

True main driver in twin's top two — **84%** of patients

The cuff carries most of the accuracy.
Wearables and MyZenn add the why — and the early behavioural warning that enables intervention.

One morning, five clicks: triage, understand, simulate, act.

A doctor starts with a risk-ranked panel, understands the why, tries what-if scenarios, and prescribes a personalized protocol through MyZenn.

1 Triage roster

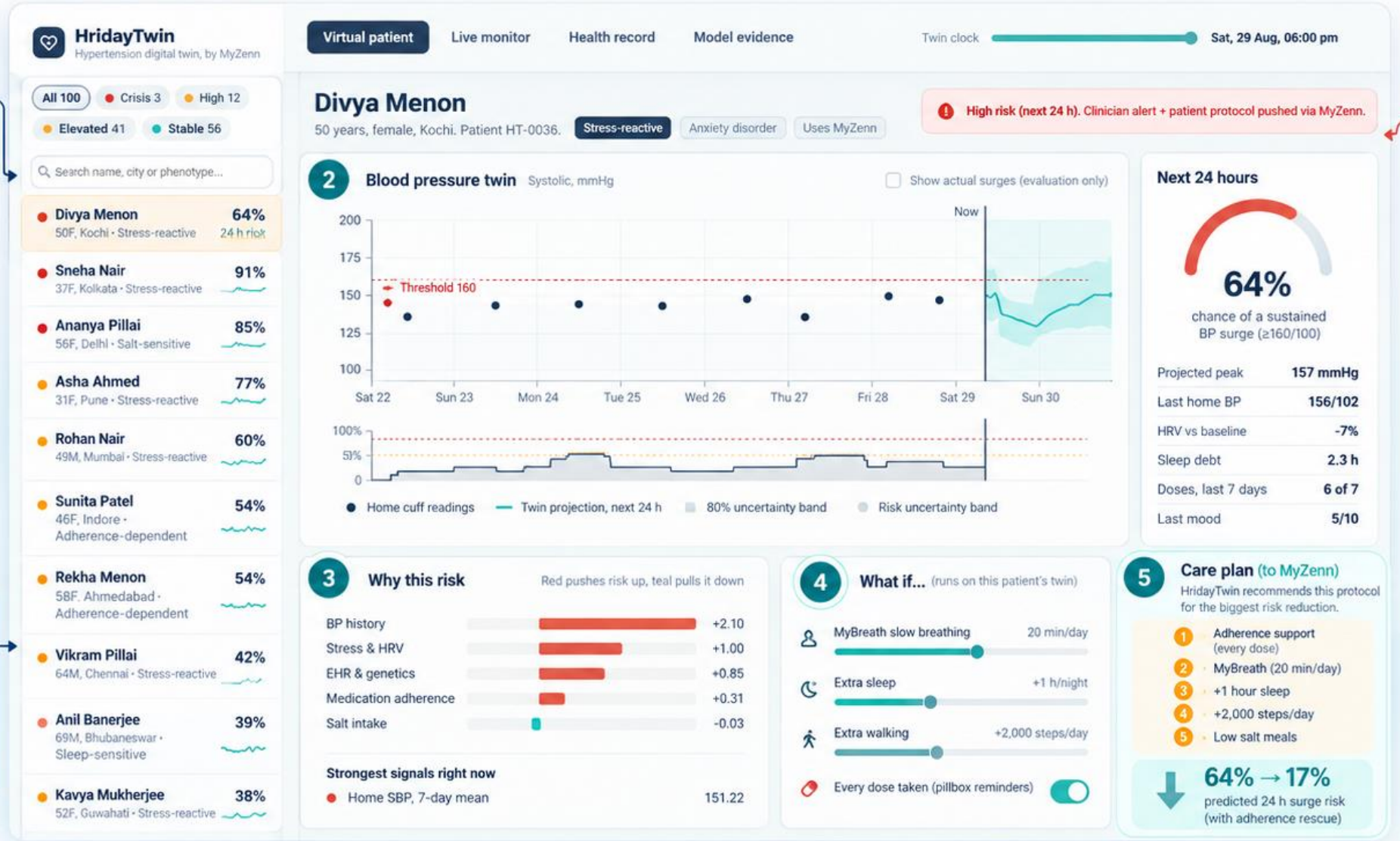
100 virtual patients ranked by 24-hour surge risk, with phenotype badges.

3 Explain today's drivers

SHAP drivers and personal sensitivities show why this patient is at risk.

Crisis alert → MySafe escalation

If a critical reading is detected, the patient is alerted and escalated.



Doctor console
Triage → Explain → Simulate → Act
Live monitor and follow up

MySentinel
Schedules protocol, logs the action, keeps learning (outcome → bandit reward)

MyZenn patient app
Guidance, nudges and practices (MyBreath / MyBody / MyYoga)

Outcome feedback
Real-world data flows back to improve the model

Live data in, the right nudge out, and the twin keeps learning.

A closed loop from real-time signals to safe, personalized intervention.

HridayTwin by MyZenn

HridayTwin — Clinician console (synthetic demo)

Virtual patient | **Live monitor** | Health record | Model evidence | Twin clock — Thu, 30 Jul, 09:00 pm

Patients (100)

Search name, city or phenotype...

All 100 | Crisis 3 | High 12 | Elevated 41 | Stable 44

- **Sanjay Iyer** — 58M, Bengaluru — Stress-reactive — 91% 24 h risk
- **Ananya Pillai** — 56F, Delhi — Sleep-sensitive — 85% 24 h risk
- **Divya Menon** — 50F, Kochi — Stress-reactive — 81% 24 h risk
- **Asha Ahmed** — 31F, Pune — Adherence-dependent — 77% 24 h risk
- **Sunita Patel** — 49F, Mumbai — Adherence-dependent — 64% 24 h risk
- **Rohan Nair** — 49M, Mumbai — Stress-reactive — 60% 24 h risk

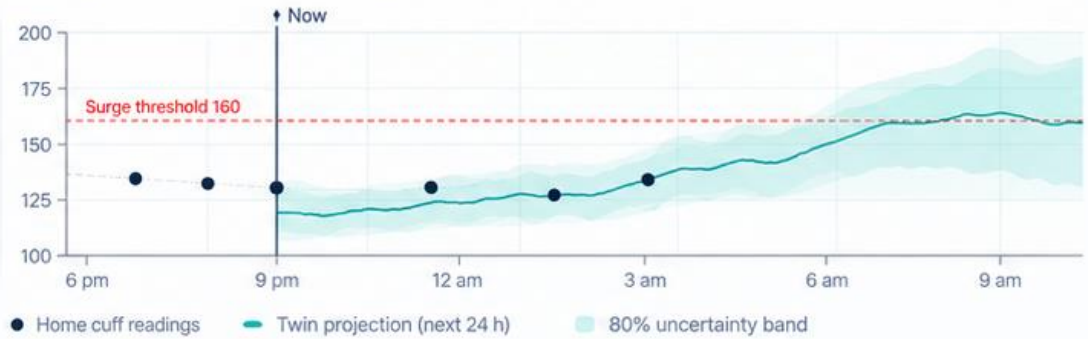
Divya Menon

50 years, female, Kochi. Patient HT-0036 (synthetic demo). Stress-reactive | Anxiety disorder | Uses MyZenn

Twin time	24 h risk	Heart rate	HRV (RMSSD)	Stress score	State
Thu, 09:00 pm	81% ↑ +12% (vs 6 h ago)	83 bpm	10 ms	68 /100	Awake

Blood pressure twin

Systolic, mmHg — Show actual surges (evaluation only)

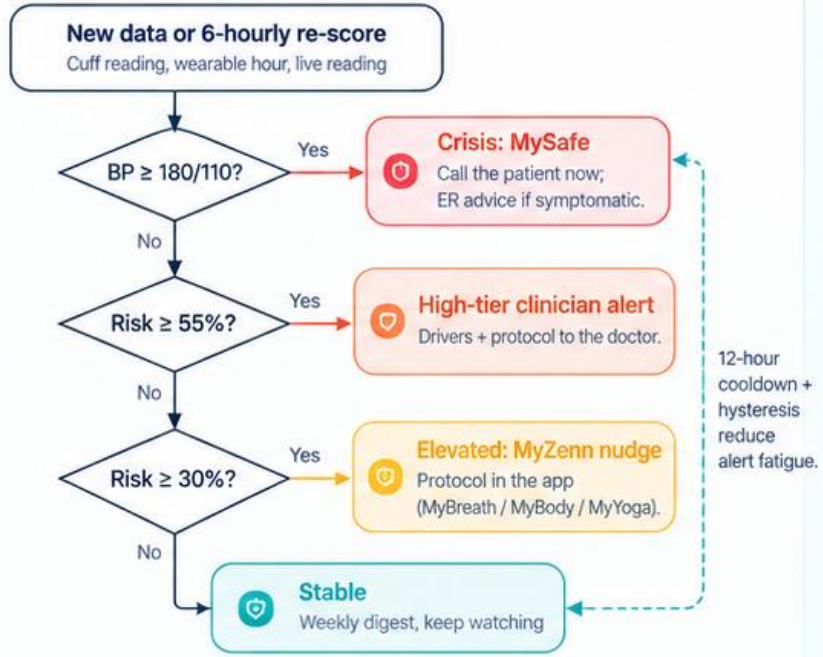


Live data sources (streaming replay)

- **WebSocket replay** — Live — 1 hour per tick (wearables, app)
- **Webcam rPPG** — Live — Video stays in browser
- **Bluetooth HR/HRV** — Live — Wearable strap or watch — HR 83 bpm, HRV 10 ms

MySentinel: from a risk score to an action

Safety rules first, then the learned policy. Thresholds from twin_engine/config.py.



LinUCB policy: picks the best protocol for this patient

Context: phenotype, drivers, current risk. Tries 5 arms: breath, sleep, adherence, salt, move.

16 points	8.5 points	18 points
average 24 h risk reduction (our policy)	random choice (baseline)	best possible choice (upper bound)

The closed loop

From real-time signals to better health days.

1. **MySentinel** — Selects and personalizes a protocol using the twin (LinUCB).
2. **MyZenn delivers the nudge** — MyBreath | MyBody | MyPeace | MyYoga
3. **Patient engages** — Practises, readings flow back, crisis readings trigger MySafe.
4. **Outcomes update the policy** — The twin keeps learning from real-world data.

Privacy and safety by design | Video never leaves the browser • No diagnosis on lock screen • Synthetic data only (no real patient data)

HridayTwin
by MyZenn

SYNTHETIC • PRIVATE • CLINICIAN-LED

Privacy, safety and clinical sense are part of the design.

Synthetic data for training. On-device processing for privacy. Clinicians stay in control.

PRIVACY

Real-world utility without real patient data.

- **Synthetic training data only**
  Synthea-style EHR, open wearables, causal simulator.
- **Camera video processed on-device**
  Only a heart-rate number leaves the device.
- **Every live reading stored with source and quality**
  Cuff, wearable, webcam, app signals.
- **DPDP-aligned consent and India data residency planned for pilot.**

SAFETY

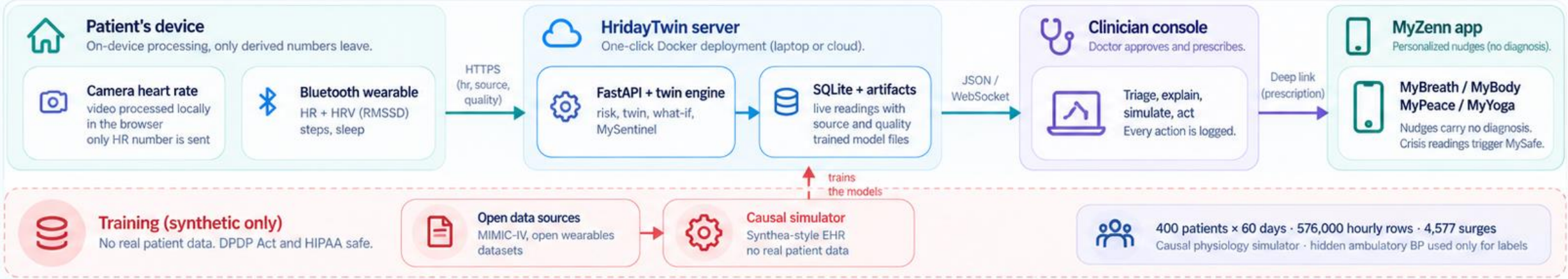
Built-in guardrails for real-world use.

- **Clinical monotonic constraints**
  e.g., more doses can never raise risk.
- **Crisis readings bypass the queue**
  ≥180/110 triggers MySafe call.
- **Doctor decides; twin supports**
  Transparent evidence, what-if, and protocol options.
- **Every action logged**
  Full audit trail from data to decision.
- **Decision support only — not a medical device.**
  No autonomous diagnosis or treatment.

HONEST LIMITS

Clear about what it can and cannot do (yet).

- **Synthetic data only so far**
  Validation on real patients is the next step.
- **Night-time BP uses a population dip assumption**
  Personalization is limited by available data.
- **Camera HR not yet validated across skin tones**
  Works well on synthetic data; real-world validation in progress.
- **Single-driver phenotype accuracy: 60%; top-two driver accuracy: 84%.**
  Some patients have multiple interacting drivers.



From prototype to patients: MyZenn is live today, and HridayTwin is the bridge to cardiometabolic care.

Predict, explain and prevent cardiometabolic surges with a digital twin that works in the real world.

Impact

Real value for doctors, patients and health systems.

For doctors

See which hypertensive patients will surge tomorrow, why, and what will stop it.

For patients

Get a gentle, specific nudge instead of a generic "reduce stress".

For health systems

One container, runs on a laptop; it scales by adding patients, not hardware.

Roadmap

From open data to real-world pilots and broader care.

1. **Retrain and validate on open data**
   MIMIC-IV, then open wearable datasets.
2. **Pilot with a partner hospital in Kolkata**
   Using ambulatory BP as ground truth.
3. **Interoperability**
   FHIR RiskAssessment export and ABDM (ABHA) linkage.
4. **Extend to type 2 diabetes**
   CGM data, using the same hybrid twin architecture.

The ask

Help us take HridayTwin from prototype to patients.

A clinical partner

- Pilot at a partner hospital (Kolkata)
- Access to ambulatory BP for ground truth
- Clinical guidance on workflow and protocols

An anonymised cohort through Happiest Health

- De-identified EHR and routine labs
- Support with ethics and governance
- Path to real-world validation and scale

Let's build this together

Clinical partner and anonymised cohort access through Happiest Health.

MyZenn is live today

MyBreath · MyBody · MyPeace · MyYoga

Personalised, on-device nudges. No diagnosis on the lock screen.

Guidance and behaviour change →

HridayTwin

The bridge to cardiometabolic care

→ Earlier, preventable events

Healthier tomorrows

- Fewer hypertensive surges
- Better cardiometabolic control
- More time in healthy days